

Designing synchronizer module in CMMS software based on lean smart maintenance and process mining

Synchronizer
module in
CMMS
software

509

Seyed Hesam Hosseinzadeh Mazloui, Alireza Moini and
Mehrdad Agha Mohammad Ali Kermani

*Faculty of Management, Economic and Progress Engineering,
Iran University of Science and Technology, Tehran, Islamic Republic of Iran*

Received 17 October 2021
Revised 26 February 2022
Accepted 24 April 2022

Abstract

Purpose – New maintenance hypotheses such as lean smart maintenance emphasized internal integration. Since the maintenance process is not fully integrated with other business processes, it indicates that some of the problems in the maintenance process are caused by other departments. Additionally, nothing can be managed or improved without first measuring it. In order to enhance internal integration, this study developed a model that makes use of information systems data to examine synchronization and collaboration across departments engaged in maintenance operations.

Design/methodology/approach – This research connects maintenance management and business process management through information systems. A conceptual module model based on CMMS is proposed that will use data which are already available in CMMS and, using process mining, will assess the level of synchronization between departments within an organization.

Findings – This conceptual model will serve as a roadmap for creating better value-added CMMS software. This system operates as a performance measurement tool in three majors, including organizational analysis, workflow analysis and eventually, a future simulation of maintenance processes. This module will serve as a decision support system, highlighting opportunities for improvement in maintenance processes.

Originality/value – A practical guideline is provided for the future development of CMMSs and their enhancement to intelligence. All assumptions are based on maintenance theories, techniques for measuring maintenance performance and business process management and process mining.

Keywords Maintenance, Smart maintenance, Lean smart maintenance, Maintenance performance measurement, Performance measurement, Information systems, CMMS, EAM, Synchronization, Process mining

Paper type Conceptual paper

Introduction

The state-of-the-art technologies of the fourth Industrial Revolution are poised to transform many of our everyday activities. The cyber-physical systems improved the intelligence and responsiveness of the equipment (Pfeiffer, 2017; Schwab, 2017; Vaidya *et al.*, 2018). Devices have become increasingly complex as a result of Industry 4.0 (I4.0) technologies, and in this competitive climate, organizations need to remain in the best possible condition (Schmiedbauer *et al.*, 2020).

A main priority is ensuring that physical assets remain in good condition, and to do so, the maintenance methods must change along with I4.0 (Srivastava *et al.*, 2018). Lean smart maintenance (LSM) is a relatively recent maintenance technique that focuses on the process of maintaining equipment (Biedermann and Kinz, 2019). Data-driven decision-making, human capital resource management, internal integration and external integration were the four



Journal of Quality in Maintenance
Engineering
Vol. 29 No. 2, 2023
pp. 509-529

© Emerald Publishing Limited
1355-2511

DOI 10.1108/JQME-10-2021-0077

The authors would like to pay special regards to Behfalab as a process mining tool vendor for funding support for this research under award number R01GM211022.

components of the smart maintenance concept. The internal integration is also concerned with internal procedures and synchronization across departments in order to complete the maintenance operations as efficiently and effectively as possible (Bokrantz *et al.*, 2020). In terms of synchronization between departments' activities, the most significant challenge is that they are handled and planned independently (Aarab *et al.*, 2017).

According to Bokrantz *et al.* (2020) research, many interviewees expressed concern over a lack of departmental synchronization. In addition, according to a study of airport cooperative research program (Delong, 2013), employees at international airports perceived a greater sense of disorganization than senior managers. Ensuring departmental synchronization helps the maintenance department work more efficiently and effectively to create more value. However, if something is not measured, it cannot be controlled (School, 2009). So there is a question: Are there any instruments available to evaluate the level of synchronization in the maintenance process?

Current maintenance performance measurement (MPM) techniques include the corporate level and indications for top-level decision-makers (Suzuki, 1994; Wireman, 2005). In addition, some research provides a mathematical model for internal integration to be used in practice (Zhai *et al.*, 2019). These approaches and models may not effectively measure synchronization; therefore, it is necessary to develop methods that enhance synchronization measurements and management. As digitalization becomes more widespread, the pace of data creation increases and in this area, E-maintenance technologies and information systems are responsible for collecting and storing data. In addition, information systems such as computerized maintenance management systems (CMMS) oversee performance measurement based on gathered data (Candell *et al.*, 2009; Bokrantz *et al.*, 2020). Accordingly, a tool for analyzing internal synchronization across departments should be included in the CMMS. Existing techniques and CMMS require simulating every organization and defining unique indicators for analyzing synchronization and collaboration level. This approach necessitates the expenditure of energy and money.

Thus, the purpose of this article is to establish a method that analyzes interdepartmental communication and coordination without the need for any extra effort and data gathering. This tool aids in the identification of issues in the maintenance processes that are linked to the collaboration of the other departments or in the maintenance department. These points are generally invisible, but they may have an impact on the performance of the maintenance department, which in turn can have an impact on the whole organization's value creation chain (Wireman, 2014).

This paper proposes a new module integrated into current CMMSs. This module does not need any previous simulation or the definition of unique indicators. It can show the level of departmental synchronization and detect fragmentation in maintenance operations. Process mining algorithms are used in conjunction with CMMS data to automatically simulate organizational processes and departmental relationships and identify and diagnose any issues that may arise (Reitemeyer, 2020). As a bonus, based on necessary periodic maintenance activities and results from process mining tools, this module can simulate the future and enhance the accuracy of scheduling activities even when there are insufficient resources available in the future.

The remaining sections of the paper are organized in the following ways. The next section reviewed the literature on synchronization, MPM, and CMMS software. Afterward, the methodology is explained, and the following section presents the conceptual module model. This module is compared to other methods in the discussion section, and successful process mining practices and the module's shortcomings are discussed. Finally, the research is concluded, and then there are references.

Literature review

Throughout this literature review, we looked at studies that have been conducted to investigate the synchronization and cooperation between the maintenance department and other departments within an organization. Thus, we needed to look at existing optimization planning studies and performance measurement frameworks to see if they could provide us a way to assess and improve the collaboration. In addition, we explored CMMSs to determine whether or not they provide useful tools for analyzing synchronization.

Synchronazation and collaboration

The definitions of synchronization are at two levels: the system level and the individual components' level. At the system level, synchronization describes how each component's dynamics happen in the same general time period. To the level of individual elements, synchronization may be thought of as a process of mutual impact, with consistent signals coming at a given element from neighboring components (Nowak *et al.*, 2017). In its most basic definition, collaboration is nothing more than a group of people working together to achieve mutual goals (Delong, 2013).

Organizational process asset management in LSM employs many management and operational activities to reduce individual and complicated steps. This makes it possible to bring together unique and separate operations in time and place to form one workflow. To be effective in I4.0, procedures need to be more flexible and networked. Digitalization aims to allow for completely autonomous process planning and control, in addition to having accurate data at your fingertips, an augmented reality technology and network optimization. There are several things to look at when you intend to automate your operations: real-time planning, automation, process traceability and simulation visualization tools. To some extent, maturity in an organization is associated with complete business process digitization, including processes in place for service systems and data lifecycle management (Schmiedbauer *et al.*, 2020; Bokrantz *et al.*, 2020).

Many research studies have been conducted on the synchronization of the maintenance department from a variety of perspectives and across a wide range of disciplines. An essential element of maintenance management that must be considered is the integration of knowledge, which is a major problem that has a considerable impact on enhancing the overall performance of industrial systems (Ruiz *et al.*, 2013). The studies of Beheshti-fakher *et al.* (2016), Aguirre and Papageorgiou (2018), Omar and Shaik (2019) and Li *et al.* (2021) suggested an operation research optimization model to combine maintenance and operation processes in order to enhance the overall performance of organizations in a variety of domains and situations. Several methods were employed to solve their problem, including genetic algorithms (Ladj *et al.*, 2016), heuristic algorithms (Guiras *et al.*, 2018) and non-linear planning (Khatab *et al.*, 2020). According to these papers, the most important issue is that the enterprises handled and planned various internal procedures independently (Aarab *et al.*, 2017; Zhai *et al.*, 2019).

Traditional business structures have a functional focus. Authority flows upward with a functional emphasis, and subordinates report to superiors. Process-oriented organizations emphasize processes over hierarchies. The focus is on outcomes and client happiness rather than inventory management or order processing processes. In determining the appropriate kind of business orientation, factors such as size, scope, and objectives should be considered (Dumas *et al.*, 2013). The process of maintenance maintains high levels of production quality. The maintenance process consists of planned and unplanned actions to maintain a physical asset in excellent operating condition. It aims to improve the cost-effectiveness of a manufacturing plant, equipment or building by creating value (Mostafa *et al.*, 2015). One of

the physical asset management principles is to ensure that the asset management system integrates with the organization's business operations processes (ISO, 2014).

Maintenance performance measurement (MPM)

MPM is the comprehensive procedure for tracking and explaining the value produced by maintenance processes to shareholders and stakeholders while providing a strategic view from the big picture viewpoint. Organizations can use MPM to understand the value created by maintenance, revise their maintenance policies and techniques and justify investment in new technologies or trends while also adjusting resource allocations. MPM also helps organizations understand the impact of maintenance on other functions and stakeholders, health and safety, among other things (Parida and Kumar, 2006; Parida and Chattopadhyay, 2007; Kumar *et al.*, 2013). Using performance measures correctly can uncover possibilities for improvement, identify issues and develop solutions (Kumar *et al.*, 2013).

MPM literature had been reviewed several times also interviews and questioner have been taken in different industries. Reviewed papers provide a big picture of MPM, for instance, its definition and boundaries (Parida and Kumar, 2009; Kumar *et al.*, 2013), challenges and issues (Parida and Kumar, 2006), development of maintenance performance indicators (MPI) and maintenance performance frameworks (Parida *et al.*, 2015). In addition, several questionnaires have been designed to identify the most utilized indicators in their respective fields (Muchiri *et al.*, 2010; Simões *et al.*, 2011; Ku and Kim, 2019; Darestani *et al.*, 2020).

A better understanding of MPIs and performance measurement frameworks is needed to answer the research question to find indicators that identify possibilities for improvement in synchronization and collaboration. The corporate level of performance measurement is the highest in the multi-level performance measurement of wireman indicator classification. Corporate performance metrics can determine what is most important to company leadership to achieve their corporate goals. It's crucial that every department in the organization operate as a team to meet their company's key performance indicators (KPI) goals. Indicators will vary for each company since they reflect market position, corporate condition and profitability. Because corporate indicators support the long-term goal, all following indicators must support the corporate strategy (Wireman, 2005). For example, overall equipment effectiveness (OEE) is a corporate-level measure that varies from industry to industry. It is a three-part analytical method for evaluating equipment performance based on its availability, performance and rate of output quality (Muchiri and Pintelon, 2008).

On the contrary to the OEE indicator, which is a high-level indicator, a low-level indicator called work order aging will be used to indicate faults in the maintenance process (SMRP, 2017). This metric calculates the age of active work orders by comparing the creation date of the work order to the current date. The result is the work order age, given in days. On the other hand, work order cycle time is defined as the amount of time it takes from the creation of a work order until it is closed in the maintenance management system (SMRP, 2017). Another type of MPIs that may determine the level of synchronization is managers' indicators in the total productive maintenance method, which generally focuses on human factors (Suzuki, 1994).

These indicators do not explain the rationale for the length of time it takes to complete a work order or the cause for a delay, nor do they indicate which step of the work order is delayed. Additionally, work orders do not cover the whole maintenance process, and human factors just focus on maintenance crew performance. Consequently, there is a need for the root cause analysis technique to explain the reasons for delays and faults (Andersen and Fagerhaug, 2006).

Because of the diverse data types produced in maintenance settings, data-driven methods for analytical maintenance assistance may be expanded beyond the equipment level. Process

mining is one of these methods (Horn and Zschech, 2019). Process mining is a new area that offers comprehensive tools to aid in the optimization of manufacturing processes (Van Der Aalst, 2016). Process mining can be used to collect data on the duration of activities, process variations, cycle time changes and the progress of degradation rates. Managers can assess the effects on each function individually using this information, which helps them optimize intervals between cost-based maintenance checks and availability functions.

Additionally, it enables a more dynamic presentation of the procedure, its results, and the ramifications for the different intervals between maintenance checks. The cyclical perspective of the proposed method enables continual modification of maintenance inspection intervals (Ruschel *et al.*, 2020). While the control-flow viewpoint emphasizes process discovery, event logs may include valuable information from various organizational, case and time perspectives. Insights about regular labor habits, organizational structures and social networks can be gained via organizational mining techniques (Van Der Aalst, 2016).

Process mining has only been utilized in a few studies in maintenance workflow. Process mining methods have been shown to assist maintenance-related goals in manufacturing environments (Horn and Zschech, 2019). The findings of several other studies looked at the actual maintenance process and showed that the information could aid decision-making (Greyling and Jooste, 2017; Thaduri *et al.*, 2019; Ruschel *et al.*, 2020). Process mining can be used from different perspectives, such as discovery, conformance and enhancement (Dakic *et al.*, 2018; Corallo *et al.*, 2020). These various views and techniques have a variety of applications in maintenance, including the detection of bottlenecks and loops and non-value-added activities and defects in the social network of workers and departments, among other things. Additionally, it will aid in the reduction of expenses and the increase of equipment uptime and the general improvement of the efficiency of maintenance. In addition, when the intervals are mined, they may be utilized for more accurate planning based on the past and for future expecting (Greyling and Jooste, 2017; Horn and Zschech, 2019; Thaduri *et al.*, 2019; Ruschel *et al.*, 2020).

Computerized maintenance management system (CMMS)

CMMS is a program created to manage the data of the equipment in business operations, providing the necessary indications for reliability improvement. It was challenging to retrieve relevant, comprehensive and consistent data from paper-based data processing systems to handle preventative maintenance without a CMMS (Braglia *et al.*, 2006). The equipment inventory, work order control and repair and maintenance history are the three major components of a CMMS (Cohen *et al.*, 2020). They are more than just a control tool for maintenance; they are also meant to enhance output quality by managing maintenance operations throughout time. In general, CMMSs are provided in a modular format to provide more flexibility and excellent interoperability across a wide range of industries (Jamkhaneh *et al.*, 2018). The scope of basic functionalities of CMMS is given below:

- (1) Administration and control of equipment.
- (2) Administration and control of the plant (purchasing cost, amortization plan, etc.).
- (3) Predictive maintenance processes (interventions, planning and scheduling).
- (4) Administration and control of drawings and documents.
- (5) Planned and unplanned maintenance management.
- (6) Work orders management.
- (7) Management of resources (technical, human).

- (8) Analysis of historical data.
- (9) Control of costs and budgets.
- (10) Inventory management and
- (11) Administration of purchase orders (Uysal and Tosun, 2012).

Efforts were made to evaluate the content of the website of many CMMS providers to see if they offer any tool for synchronization measurement. Some of the CMMS providers include UpKeep, FMX, Fiix, Maintenance Connection, MPulse, FreshService, Limble, Mobility Work, Asset panda, Axserion, TheWorkHub, Megamation Directline, QWare, ManWinWin, Maxpanda, MaintainX, Leading2Lean, Fractal, Manager Plus, IBM Maximo, Infor EAM, eMaint, Assetworks, EZOfficeinventory, ServiceChannel, FTMaintenance, SAP EAM, Oracle EAM and Asset essentials.

KPI analysis, graphical reports, configurable dashboards and scheduled reports are common features across all these applications. The Manager plus software's reports module has a log analysis feature. From another perspective, CMMS providers that also deliver other information systems, such as ERP, provide a more synchronized and integrated set of information systems, such as IBM, SAP, Oracle and Infor. However, many other CMMSs, such as Mobility Works, Maxpanda and others, have open APIs for integrating and transferring data with other information systems inside a company.

To conclude the literature review, numerous efforts have been made to propose a mathematical model for integrating and synchronizing the maintenance process with other business processes. Some surveys have addressed this issue. On the other hand, MPM frameworks and CMMS reporting tools are oriented toward financial analysis and do not adequately address synchronization analysis. This problem requires a simple tool that does not require excessive effort to evaluate level synchronization to identify bottlenecks and improve efficiency and effectiveness of maintenance department performance.

Methodology

This study is intended to offer an empirical framework for addressing the problem of departmental relations flaws with the maintenance department. This research attempted to link maintenance integration, MPM, information systems and process mining. To reconcile these, a conceptual module model for information systems is designed (Jaakkola, 2020).

The methodology of designing this conceptual module model consists of three phases. First, an investigation has been made in maintenance methods and performance measurement methods and indicators to find a solution for improving the integration and synchronization of maintenance within the organization. Second, we searched for techniques to evaluate the synchronization level in this phase, which found process mining as the best solution. In addition, after nominating the process mining, its requirements were reviewed and presented as the assumption in the following. Third, this phase includes forming the theories found in the two other steps into a conceptual model. The module model is proposed using the unified modeling language (UML) diagrams. UML is a graphical modeling language that is used to define, visualize, build and document systems. The UML has been successfully utilized to represent a variety of complex systems and has been included in various e-maintenance studies (Campos *et al.*, 2010; Espíndola *et al.*, 2013, López-campos *et al.*, 2013).

Use-case diagrams describe the behavior of a system and assist in capturing the needs of the system. Use-case diagrams illustrate the high-level functionality and scope of a system. These diagrams also highlight the interactions between the system and its actors. The use cases and actors in use-case diagrams explain what the system performs and how the actors utilize it. Class diagrams represent the items that make up the system, show the connections

between the objects, and explain what those objects perform and the services they offer (Fowler, 2004; Pilone and Pitman, 2005).

Module model

This module is described in four perspectives: functionality, inputs, algorithms and outputs. These four perspectives clearly explained the elements and functions of the module.

Functionality

We used a use-case diagram to define the synchronizer module to better explain its service and user interaction. The interaction between actors and use cases is shown in Figure 1.

This module consisted of three major parts, inputs, process mining and outputs. Use cases include:

- (1) Event log generation
- (2) Data qualification
- (3) Organizational perspective (analysis)
- (4) Social network (analysis)
- (5) Without failure report (workflow analysis)
- (6) With failure report (workflow analysis)
- (7) Future resource analysis
- (8) Structure analysis

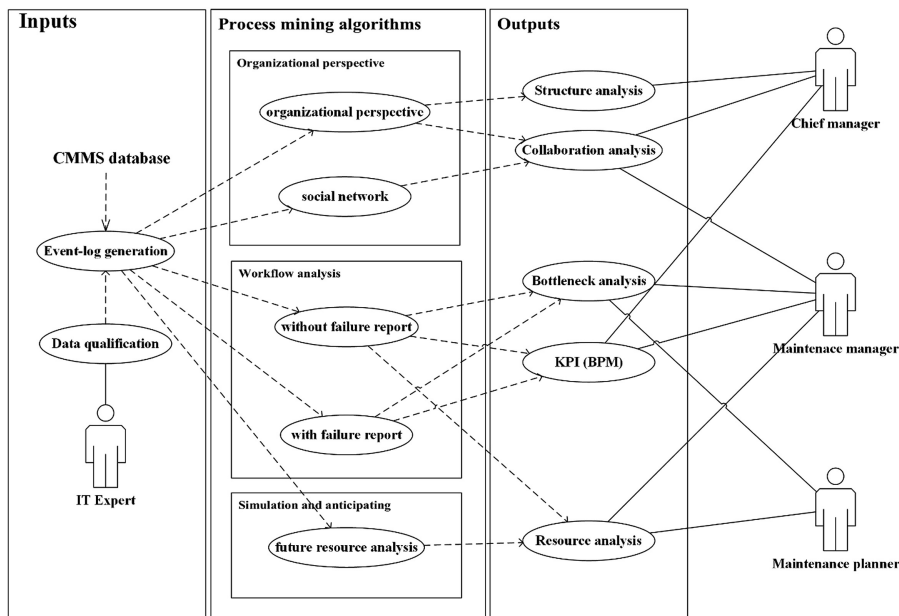


Figure 1.
Use-case diagram of
the synchronizer
module

- (9) Collaboration analysis
- (10) Bottleneck analysis
- (11) KPI (based on BPM)
- (12) Resource analysis

Actors:

- (1) IT-expert
- (2) Chief manager
- (3) Maintenance manager
- (4) Maintenance planner

In the input part, data enters from CMMS database and with a query, it turns into event logs. Event-log will be qualified by an information technology (IT) expert based on data quality guidelines explained in the next section. Data will be analyzed from three different perspectives in the process mining subsection. From the organizational perspective, the activity handover of departments and their relationships and the social network of employees who play a role in the maintenance process can be mined for insights. The maintenance workflow can be derived from two different data sets: first, maintenance processes that began with a regular periodic maintenance request. Second, maintenance procedures that were initiated because of a failure report. The future resource simulation is based on the regular periodic process model and the meantime of activities involved in the whole process.

In the third set of use cases, the information is used by various management levels, including the chief manager, the maintenance manager and the maintenance planner. The first level of organization is concerned with the organization's structure, coordination and collaboration. The second level concentrated on the flow of maintenance work within the department, as well as the efficiency of the maintenance crew members. With the assistance of the simulation, the third level assists the maintenance planner in setting more accurate maintenance plans.

Inputs

CMMS software is an excellent source of historical data that can automatically generate conceptual enterprise models. This model creation is not restricted to a particular kind of organization and will be applicable across all industries. Process mining is one of the most acceptable methods for autonomously generating conceptual enterprise models ([Reitemeyer, 2020](#)). CMMS/Enterprise Asset Management (EAM) or any other program with historical data of maintenance activities will be used to extract data for this module, which will then be processed using process mining algorithms to serve as a performance assessment tool.

Process mining techniques need a kind of data known as event logs. Event logs capture all system events, such as adding a file or program, altering the system's settings, shutting down the system, changing the date, etc. The documentation groups events into five categories: system, security, application, directory service, DNS server and DFS replication ([Van Der Aalst, 2016](#)). Logs from information systems such as CMMS/EAM or Enterprise Resource Planning (ERP) should generate the required event log for the synchronizer module. Because it is needed to cover all digital events associated with the maintenance process, we may need to integrate various databases inside the business. This event log must cover the following information:

- (1) Asset ID

- (2) Work order
 - Generation
 - Planning
 - Assigning
 - Report receiving
- (3) Inventory
 - Part request
 - Part delivery
 - Procurement
- (4) Permissions
 - Cost related
 - Security related
- (5) Administrative correspondence
- (6) Reports
 - Related to any stage of the process
- (7) Asset failure report
 - Operator report
 - NDT [1]
 - CBM [2]/IIoT [3]
 - Statistical quality control

These are the elements that must be considered while creating an event log. In the scope of maintenance, quite a few research are log-based, and they all suggest an essentially similar format for event logs (Van Der Aalst *et al.*, 2007; Sipos *et al.*, 2014; Wang *et al.*, 2017; Horn and Zschech, 2019; Thaduri *et al.*, 2019; Gutschi *et al.*, 2019; Ruschel *et al.*, 2020). The event log structure should be as follows in Table 1 as a sample, based on mentioned elements and considering those studies, process mining requirements.

Data sets

Since this data set contains essential and comprehensive information, it may be utilized for various applications. Thus, we divide this data set into three plus one data sets, as shown in Figure 2 in the notation of a UML class diagram.

Case ID (asset id)	Timestamp	Activity name	Resource (person/department)
001	2021-01-24 17:49:44.537	S1	A7/S
002	2021-01-24 15:09:35.970	S1	A7/S
001	2021-01-25 03:55:56.872	S2	A8/S
002	2021-01-27 00:14:42.463	S3	B6/M
002	2021-01-27 08:10:25.470	S7	U9/I

Table 1.
Event-log structure

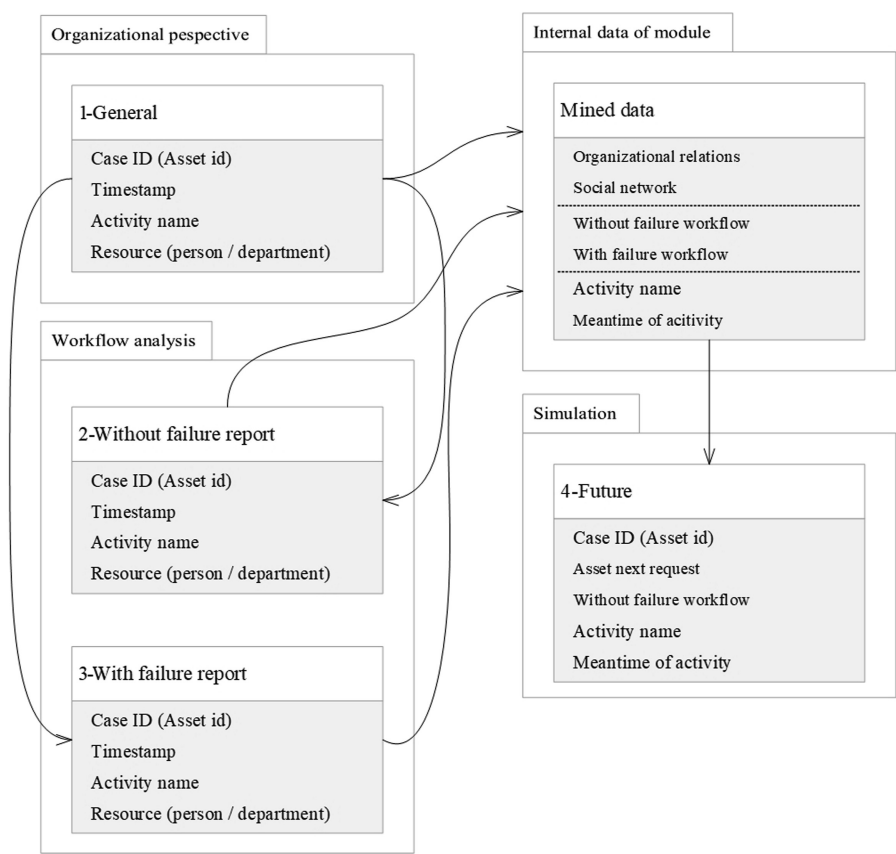


Figure 2.
Module class diagram
(data sets)

(1) General

This data set is equal to the entire data set.

(2) Without failure report

A subset of the general data set contains event data for all maintenance procedures that do not begin with a failure report and is taken from the overall data set.

(3) With failure report

Another subset from the general data set includes maintenance processes data that begins with a failure report.

(4) Future

An artificial data set will be produced based on the information obtained and stored in the internal data set of the module (Choueiri *et al.*, 2020). Based on the workflow model and future asset work requests and the meantime of each activity, this data set will be generated.

Since planned activities have varying levels of priority, speed, and budget, the behavior of companies in reactive and proactive maintenance procedures is distinct. Accordingly, we

separate these data sets in order to prevent bias in the results. Besides, there can be other criteria to make new data sets; for example, maintenance processes can be divided based on assets priority.

Data qualification

These four data sets should be qualified based on the following assumptions and criteria:

- (1) Assumptions
 - Suitable for large and medium companies
 - Suitable for long-term and mid-term analysis
 - Continuously recorded data needed
 - Structured data needed
- (2) Criteria

Data qualification should meet both CMMS and process mining criteria.

- (1) CMMS
 - Sampling size (at least 100 cycles)
 - Consistency
 - Free-of-error
 - Completeness
 - Plausibility ([Gitzel et al., 2018](#))
- (2) Process mining
 - No Missing data

Missing event or partial or incomplete trace.

- No timestamp errors
- No ambiguity between events

Duplicate events, overlapping activity executions.

- Flexible processes need to be clustered into subsets ([Bose et al., 2013](#); [Wynn and Sadiq, 2019](#)).

Algorithms and tools

Algorithms. There are many ways to build a process model using event logs. Discovery: This method creates a process model using the process systems' produced event log. Existing systems can detect actual processes from event logs based on simulations at different process phases. Conformance: The created process model from discovery is evaluated with fresh event logs generated from the same process for conformance model correctness. Procedural, organizational and declarative methods of conformance checking may be used in corporate rules/policies, legislation, etc. Enhancement: This phase modifies or transforms the previously established process model based on data from the event log. The fundamental distinction

between conformance testing and enhancement is that the former checks model-reality alignment while the latter modifies or advances the established model (Van Der Aalst, 2016).

Process mining uses algorithms to mine data according to rules. Each algorithm has pros and cons. Data mining algorithms fall into three categories: deterministic, heuristic and genetic (Van Der Aalst, 2016; Greyling and Jooste, 2017). With deterministic models, all the data required to determine the result of the process mining is known. One of the most striking features of deterministic models is that the output of process mining is constant for the given event log input variables (Greyling and Jooste, 2017). The α -algorithm is a deterministic algorithm concerned with the ordering and relational connections of events in an event log (Van Der Aalst, 2016).

When a predefined algorithmic method cannot discover an optimum solution, heuristic algorithms are often employed. In these instances, it is essential to use a trial-and-error form to identify a suitable solution (whether globally optimum or not). Process mining is based on deterministic algorithms, supplemented with Pareto principles (frequency suggestive of significance) to allow for the removal of routes or events that add unneeded complexity. Genetic algorithms are predicated on the basic principles of evolutionary theory that control natural selection. A solution to a problem is discovered using genetic algorithms by beginning with an arbitrary starting point and then looking for a better answer while discarding inferior alternatives (Van Der Aalst, 2016; Greyling and Jooste, 2017; Corallo *et al.*, 2020).

In addition, there are organizational perspective algorithms. Providing support for collaboration in workflow systems and modeling organizational information in the context of a business. Some approaches examine resource impact on process performance. In other cases, the underlying organizational model or social network is discovered by extracting static relations between process participants (e.g. delegation). Using enriched event logs, advanced research can extract dynamic relations regarding actual communication between process participants (Schönig *et al.*, 2016, 2018).

Tools. There are several process mining tools available today. Some of them are appropriate for commercial purposes, while others are appropriate for academic study. According to the reviewed article by Corallo *et al.* (2020), ProM is a dominating instrument in academic research, and Disco has also been utilized for academic purposes. Table 2 illustrates process mining tools.

After introducing algorithms and tools, specific types of algorithms are selected for each data set.

(1) General

Organizational perspective algorithms are selected for analyzing this data set. Organization relations and social networks can be mined in this step. The ProM, Rapid ProM and My Invenio can be used.

Table 2.
Process mining tools

Name	Description
ProM	Academic, with +600 add ups
Rapid ProM	Academic, power full in analysis and many add ups
Disco	Commercial, need installation
Celonis	Commercial, SaaS
My Invenio	Commercial, need installation
Minit	Commercial
QPR	Commercial
Lexmark	Commercial
BehfaLab	Commercial, SaaS, the first and only Iranian process mining tool

(2) Without failure and with failure data set

Workflow mining algorithms are selected to analyze these two data sets to make a process model, conformance check and mining meantime of each activity. All the academic and commercial tools are able to perform this.

(3) Future simulation

All the mentioned tools can simulate and show what will happen in future by animations.

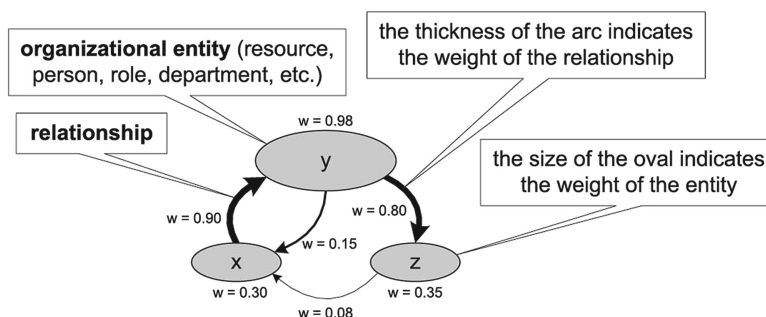
521*Outputs*

The module's outputs are divided into three categories: organizational, workflow and future simulation.

Organizational. The primary goal of this study was to assess the level of collaboration and synchronization across the department in the maintenance process. The organizational outputs of this module will include a model of departmental relationships as well as a quantification of the synchronization. A departmental relationships sample model is shown in Figure 3.

The use of social network mining can aid in the completion of the synchronization evaluation. With this output, the handover of work and the level of collaboration can be evaluated. It is possible to visualize the social network, and it is possible to analyze it using social network analysis tools to determine each employee's role in the maintenance processes. Figure 4 illustrates a sample of a social network in a company they are cooperating in a process. In addition, a matrix will be produced to quantify the handovers, and it will serve as an indication that can be used to compare the handovers with other periods in the future; Table 3 is a sample matrix related to Figure 4's network (names in Figure 4 and Table 3 are example). Additionally, it is feasible to detect an overloaded workload at a particular job inside a company, and it is conceivable that an additional employee will be required.

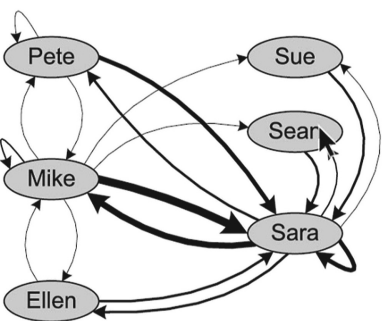
Workflow. The workflow outputs will demonstrate how processes really take place in the organization. The module will visualize the process, statistics and performance metrics. First, visualizer tools such as the dependency graph (Figure 5) and Business Process Model and Notation (BPMN) are used to depict the actual process. Bottlenecks, loops and non-value-added activities, as well as any other fragments, can be identified with the assistance of these models and graphs. This graph visualizes the path and order of actions within the process. In addition, using a basis process model or previous process mining results enable us to do a conformance check to ensure that the actual process is consistent with the basis process model.



Note(s): Van Der Aalst, 2016

Figure 3.
Sample of the
departmental
relationship
(organizational
perspective in process
mining)

Figure 4.
Sample of the social
network of an
organization



Note(s): Van Der Aalst, 2016

Table 3.
Sample of the activity
handover matrix

	Pete	Mike	Ellen	Sue	Sean	Sara
Pete	0.135	0.225	0.09	0.06	0.09	1.035
Mike	0.225	0.375	0.15	0.1	0.15	1.725
Ellen	0.09	0.15	0.06	0.04	0.06	0.69
Sue	0	0	0	0	0	0.46
Sean	0	0	0	0	0	0.69
Sara	0.885	1.475	0.59	0.26	0.39	1.3

Source(s): Van Der Aalst (2016)

There is a tool known as the Dotted graph (Figure 6). This graph depicts the queues that occurred at each step of the process; this information is helpful for decision-makers to minimize the time required for maintenance operations. As a result, the module notifies the maintenance crew of both process model problems and task load concerns.

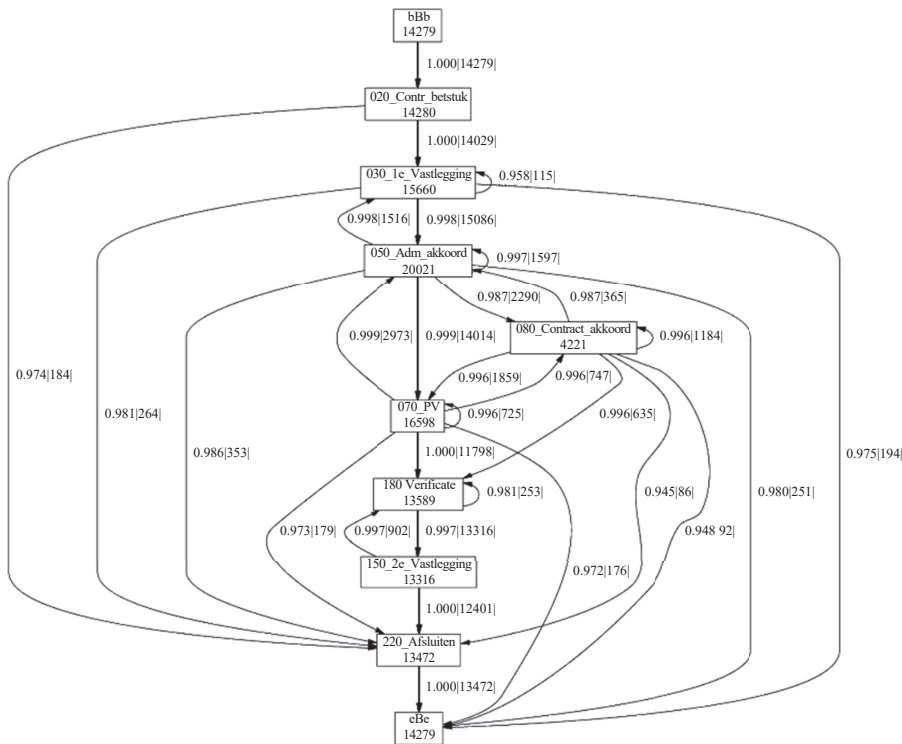
Quantitative analysis tools, on the other hand, are used to analyze process results. For example, the histogram of activity time is provided, which is very useful in identifying issues. In addition, there are BPM performance indicators that can be utilized to make comparisons across organizations. In general, these tools enable us to understand better the performance of the maintenance team and everyone else involved in the maintenance process.

As mentioned previously, it is possible to focus on specific assets data, and this module has the ability to track the trace of them. For example, assets with critical priority or assets with an issue in RAMS indicators can be followed by the sequence of activities of their maintenance procedure.

Future simulation. Future simulation allows decision-makers to visualize what will happen if everything continues in the same manner as in the past. They can plan in a more synchronized manner with other departments and procedures. Additionally, they can manage the required human resources or other required resources. Maintenance planners, for example, can quickly identify the need for outsourcing and can handle the administrative processes associated with it.

Discussion

Upon reviewing MPM frameworks and performance indicators, a few tools can support the synchronization level assessment; nevertheless, in this part, we compared these tools to the suggested module model. There are various indicators in MPM frameworks focused on the efficiency and effectiveness of maintenance, ranging from high-level indicators such as



Note(s): Van Der Aalst *et al.* (2007)

Figure 5.
Sample of the
dependency graph

OEE to more specific indicators such as work order age (SMRP, 2017). Financial metrics received the majority of focus from frameworks, with internal processes receiving less attention (Kumar *et al.*, 2013). Overall, there were unsatisfactory attempts to improve the internal procedures associated with the equipment's maintenance. In this research, we attempted to offer a module model focused on internal processes and departmental relationships and anything that impacts maintenance performance related to other departments.

The most notable feature of this module is that it does not require extensive modification in various organizations and is not restricted to a particular industry or organization. It is a performance measuring tool that can enhance internal operations. It also provides information about activities in real-time, which can be used to plan more accurately and synchronously. In other words, this module can assist decision-makers in developing data-driven plans rather than just relying on experience.

Other studies have adopted the integrated maintenance and operation approach to creating a model based on operation research to synchronize maintenance with other parts of a company's operations (Beheshti-fakher *et al.*, 2016; Aguirre and Papageorgiou, 2018; Omar and Shaik, 2019; Li *et al.*, 2021). These optimization issues are only applicable to a single business in a particular sector. To implement such an optimization model in another business, they must first complete a project that includes data collection, objective definition and other requirements. In the model discussed in this paper, the current data were examined without any further effort, and the company's performance was evaluated on a regular basis by this

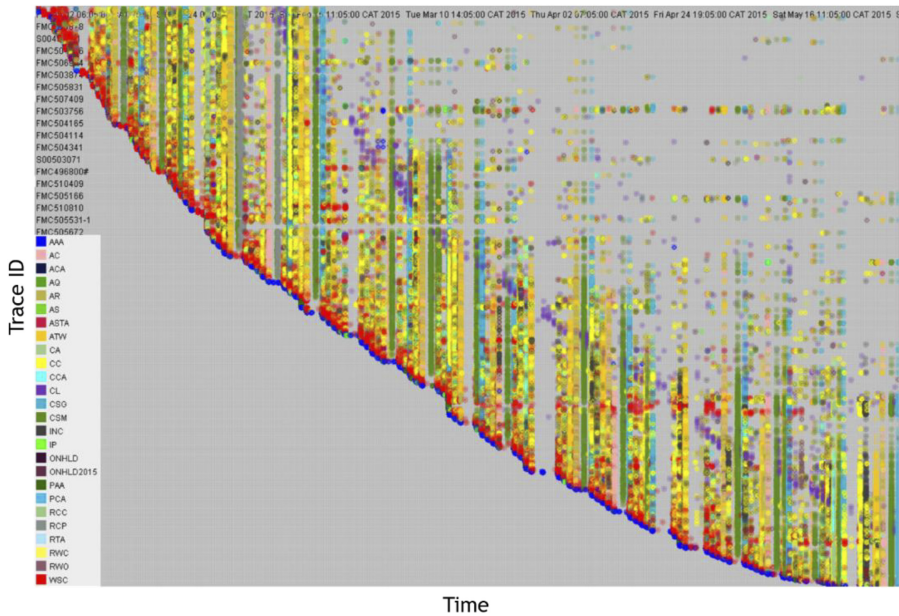


Figure 6.
Sample of the
dotted chart

Source(s): Greyling and Jooste (2017)

module. This module is a great tool to assist maintenance planners in setting the best maintenance plans.

A further point of view is that this module impacts organizational culture in certain ways. First and mainly, it serves to demonstrate the relevance of other departments in the maintenance process, as well as the function of maintenance and its importance, which is an excellent step toward becoming a process-oriented organization rather than a functional-oriented organization. In a process-oriented organization, different departments collaborate to achieve common objectives. Businesses with this attitude encourage collaboration across production, sales and other divisions to discover methods and improve their operations (Dumas *et al.*, 2013).

Furthermore, many people feel that CMMS/EAM or similar information systems are not sufficiently useful and do not contribute value to the company's value chain. Businesses management does not take them seriously, which is one of the primary causes for the failure of CMMS implementation projects (Wireman, 2008). This process mining technique and demonstration of its function in a process model may be used to emphasize the importance of CMMSs (by extension, the maintenance department) and alter people's perceptions of them. In addition, the synchronizer module can aid in the increased involvement of information systems and maintenance procedures that will take place.

Successful process mining techniques have been carried out in the context of maintenance, and the outcomes of process mining in maintenance processes have been remarkably successful. For example, [Greyling and Jooste \(2017\)](#) used the process mining method in a Petro-chemical organization, a production company. They concentrated on the traces and process model within the business. They identified loops, compliance and deviations in the process. The findings were unexpected; only 11.53% of traces were planned. They also utilized KPI and dotted charts to evaluate operations and their meantime. Additional studies

in production-type businesses utilized process mining to enhance and integrate the maintenance with output. [Ruschel et al. \(2020\)](#) used fuzzy process mining and Bayesian networks in their study to determine the optimum maintenance intervals, process behavior and cost-related problems that are synced with production operations. [Horn and Zschech \(2019\)](#) used these methods, and in addition, they considered organizational perspective by mining the social network of maintenance process inside a car tire manufacturing business. A framework based on the factory information system and production data were suggested by [Kurscheidt et al. \(2021\)](#) as a decision support system to assess the probability of failure and the requirement for maintenance.

On the other hand, [Thaduri et al. \(2019\)](#) utilized process mining to assess maintenance procedures in the iron-ore track railway in the north of Sweden. They focused on three essential variables, process bottleneck, traffic disturbance and asset systems which indicate that their research is a mix of fleet and asset-based maintenance. They considered administrative, logistical, maintenance activities and repair delays while analyzing the processes. They also utilized the collected information for future prediction in a decision support system. Many additional studies have utilized process mining in the industry with a diverse variety of applications, as reviewed by [Corallo et al. \(2020\)](#). These examples demonstrate the capability and feasibility of the process mining and the synchronizer module in information systems as a decision-making assistance tool.

A few points regarding the module discussed in this paper need to be explained. First and foremost, its name; we named it the synchronizer, this module identifies flaws and possibilities for improvement, but it does not mean that it can correct them without any effort. Rather than that, decision-makers should consider the module's output and decide according to business process management standards ([Dumas et al., 2013](#)). The second critical point is that this module is not intended to replace existing MPM frameworks and KPIs. In fact, it is a complement to the standard performance measurement tools. A performance indicator must meet six requirements to be classified as a KPI, including being numerical and linear, which this module does not meet and is not classified as a KPI ([Roubtsova and Michell, 2013](#)). It is also necessary to clarify the limits of this module. For starters, the quality of the data is critical for this module to achieve complete analysis. Secondly, it is ineffective when used for a short-term evaluation. In addition, it is not beneficial to small businesses since their maintenance processes are not too complex. There are other capabilities of process mining, but for the purposes of this module, we assumed that there is regular existing data in an organization and that it can function fully automatically and does not need further effort.

Conclusion

In this research, we discussed a gap that had been discovered by previous investigations, and we discussed how certain theories emphasized controlling this gap. According to this gap, there are flaws in the internal integration of the maintenance process with other business processes, indicating that some of the issues in the maintenance process are linked to other departments. Furthermore, it is impossible to manage and enhance anything if you do not measure it first. We suggested an attachable module to information systems to evaluate and display the degree of synchronization and cooperation between various departments within maintenance processes, using information systems data.

This module uses existing data stored in information systems and employs a technique known as process mining. Process mining is a strong tool that is being utilized in many successful practices across a wide range of industries and businesses. A significant portion of the output from this module is organizational analysis, followed by workflow analysis, and finally, the future simulation. This module operates as a decision support system and identifies opportunities for improvement within the maintenance procedures.

As part of our ongoing research, we will attempt to make information systems smarter by adding additional modules that provide new capabilities and differentiate them from a basic excel table. These kinds of research will develop information systems, making them more intelligent, user-friendlier and allowing them to become more widespread.

Notes

1. Nondestructive Test
2. Condition based monitoring
3. Industrial Internet of things

References

- Aguirre, A.M. and Papageorgiou, L.G. (2018), "Medium-term optimization-based approach for the integration of production planning, scheduling and maintenance", *Computers and Chemical Engineering*, Vol. 116, pp. 191-211.
- Andersen, B. and Fagerhaug, T. (2006), *Root Cause Analysis: Simplified Tools and Techniques*, Quality Press, Milwaukee.
- Arab, S., Barkany, A.E. and Khalfi, A.E. (2017), "The integration of maintenance plans and production scheduling taking account of outsourcing: a literature review", *International Journal of Productivity and Quality Management*, Vol. 21, pp. 1-22.
- Beheshti-fakher, H., Nourelfath, M. and Gendreau, M. (2016), "Joint planning of production and maintenance in a single machine deteriorating system", *IFAC-PapersOnLine*, Vol. 49, pp. 745-750.
- Biedermann, H. and Kinz, A. (2019), "Lean smart maintenance—value adding, flexible, and intelligent asset management", *BHM Berg-Und Hüttenmännische Monatshefte*, Vol. 164, pp. 13-18.
- Bokrantz, J., Skoogh, A., Berlin, C., Wuest, T. and Stahre, J. (2020), "Smart Maintenance: an empirically grounded conceptualization", *International Journal of Production Economics*, Vol. 223, 107534.
- Bose, R.J.C., Mans, R.S. and Van Der Aalst, W.M. (2013), "Wanna improve process mining results?", *2013 IEEE symposium on computational intelligence and data mining (CIDM)*, pp. 127-134, IEEE.
- Braglia, M., Carmignani, G., Frosolini, M. and Grassi, A. (2006), "AHP-based evaluation of CMMS software", *Journal of Manufacturing Technology Management*, Vol. 17 No. 5, pp. 585-602.
- Campos, M.A.L., Fumagalli, L., Fernández, J.F.G., Márquez, A.C. and Macchi, M. (2010), "UML model for integration between RCM and CBM in an e-Maintenance architecture", *IFAC Proceedings Volumes*, Vol. 43, pp. 110-115.
- Candell, O., Karim, R. and Söderholm, P. (2009), "eMaintenance—information logistics for maintenance support", *Robotics and Computer-Integrated Manufacturing*, Vol. 25, pp. 937-944.
- Choueiri, A.C., Sato, D.M.V., Scalabrin, E.E. and Santos, E.A.P. (2020), "An extended model for remaining time prediction in manufacturing systems using process mining", *Journal of Manufacturing Systems*, Vol. 56, pp. 188-201.
- Cohen, T., Baretich, M.F. and Gentles, W.M. (2020), *Computerized Maintenance Management Systems. Clinical Engineering Handbook*, Academic Press, Cambridge, MA, pp. 208-218.
- Corallo, A., Lazoi, M. and Striani, F. (2020), "Process mining and industrial applications: a systematic literature review", *Knowledge and Process Management*, Vol. 27, pp. 225-233.
- Dakic, D., Stefanovic, D., Cosic, I., Lolic, T. and Medojevic, M. (2018), "Business process mining application: a literature review", *Annals of DAAAM and Proceedings*, Vol. 29, pp. 866-875.
- Darestani, S.A., Ganji, M. and Imannezhad, R. (2020), "What are the key determinants of maintenance performance?", *Production*, Vol. 30, e20190155, doi: [10.1590/0103-6513.20190155](https://doi.org/10.1590/0103-6513.20190155).

- Delong, J.C. (2013), *Guidebook to Creating a Collaborative Environment Between Airport Operations and Maintenance*, Transportation Research Board, Washington, DC, Vol. 92.
- Dumas, M., La rosa, M., Mendling, J. and Reijers, H.A. (2013), *Fundamentals of Business Process Management*, Springer, Heidelberg, Vol. 1.
- Espíndola, D.B., Fumagalli, L., Garetti, M., Pereira, C.E., Botelho, S.S. and Henriques, R.V. (2013), "A model-based approach for data integration to improve maintenance management by mixed reality", *Computers in Industry*, Vol. 64, pp. 376-391.
- Fowler, M. (2004), *UML Distilled: A Brief Guide to the Standard Object Modeling Language*, Addison-Wesley Professional, Boston, MA.
- Gitzel, R., Subbiah, S. and Ganz, C. (2018), *A Data Quality Dashboard for CMMS Data*, ICORES, Setúbal, pp. 170-177.
- Greyling, B.T. and Jooste, W. (2017), "The application of business process mining to improving a physical asset management process: a case study", *South African Journal of Industrial Engineering*, Vol. 28, pp. 120-132.
- Guiras, Z., Turki, S., Rezg, N. and Dolgui, A. (2018), "Optimization of two-level disassembly/remufacturing/assembly system with an integrated maintenance strategy", *Applied Sciences*, Vol. 8, p. 666.
- Gutschi, C., Furian, N., Suschnigg, J., Neubacher, D. and Voessner, S. (2019), "Log-based predictive maintenance in discrete parts manufacturing", *Procedia CIRP*, Vol. 79, pp. 528-533.
- Horn, R. and Zschech, P. (2019), "Application of process mining techniques to support maintenance-related objectives".
- ISO (2014), "ISO 55001:2014 asset management — management systems — requirements".
- Jaakkola, E. (2020), "Designing conceptual articles: four approaches", *AMS Review*, Vol. 10, pp. 18-26.
- Jamkhaneh, H.B., Pool, J.K., Khaksar, S.M.S., Arabzad, S.M. and Kazemi, R.V. (2018), "Impacts of computerized maintenance management system and relevant supportive organizational factors on total productive maintenance", *Benchmarking: An International Journal*, Vol. 25 No. 7, pp. 2230-2247.
- Khatab, A., Diallo, C., Aghezzaf, E.-H. and Venkatadri, U. (2020), "Optimization of the integrated fleet-level imperfect selective maintenance and repairpersons assignment problem", *Journal of Intelligent Manufacturing*, Vol. 33, pp. 703-718.
- Ku, S. and Kim, C. (2019), "Development of a model for maintenance performance measurement", *Journal of Quality in Maintenance Engineering*, Vol. 26 No. 1, pp. 69-86.
- Kumar, U., Parida, A., Duffuaa, S.O., Galar, D., Stenström, C. and Berges, L. (2013), "Maintenance performance metrics: a state-of-the-art review", *Journal of Quality in Maintenance Engineering*, Vol. 19 No. 3, pp. 233-277.
- Kurscheidt netto, R.J., Fr Loures, E.D. and Dos Santos, E.A. (2021), "Enabling the use of shop floor information for multi-criteria decision making in maintenance prediction", *International Joint conference on Industrial Engineering and Operations Management*, pp. 435-447, Springer.
- Ladj, A., Varnier, C. and Tayeb, F.B.-S. (2016), "IPro-GA: an integrated prognostic based GA for scheduling jobs and predictive maintenance in a single multifunctional machine", *IFAC-PapersOnLine*, Vol. 49, pp. 1821-1826.
- Li, L., Wang, Y. and Lin, K.-Y. (2021), "Preventive maintenance scheduling optimization based on opportunistic production-maintenance synchronization", *Journal of Intelligent Manufacturing*, Vol. 32, pp. 545-558.
- López-campos, M.A., Márquez, A.C. and Fernández, J.F.G. (2013), "Modelling using UML and BPMN the integration of open reliability, maintenance and condition monitoring management systems: an application in an electric transformer system", *Computers in Industry*, Vol. 64, pp. 524-542.

- Mostafa, S., Dumrak, J. and Soltan, H. (2015), "Lean maintenance roadmap", *Procedia Manufacturing*, Vol. 2, pp. 434-444.
- Muchiri, P. and Pintelon, L. (2008), "Performance measurement using overall equipment effectiveness (OEE): literature review and practical application discussion", *International Journal of Production Research*, Vol. 46, pp. 3517-3535.
- Muchiri, P.N., Pintelon, L., Martin, H. and De Meyer, A.-M. (2010), "Empirical analysis of maintenance performance measurement in Belgian industries", *International Journal of Production Research*, Vol. 48, pp. 5905-5924.
- Nowak, A., Vallacher, R.R., Zochowski, M. and Rychwalska, A. (2017), "Functional synchronization: the emergence of coordinated activity in human systems", *Frontiers in Psychology*, Vol. 8, p. 945.
- Omar, R. and Shaik, M.A. (2019), "Integrated scheduling of production and maintenance for continuous plants: conditional sequencing and approximate modeling of storage", *Industrial and Engineering Chemistry Research*, Vol. 58, pp. 19100-19121.
- Parida, A. and Chattopadhyay, G. (2007), "Development of a multi-criteria hierarchical framework for maintenance performance measurement (MPM)", *Journal of Quality in Maintenance Engineering*, Vol. 13 No. 3, pp. 241-258.
- Parida, A. and Kumar, U. (2006), "Maintenance performance measurement (MPM): issues and challenges", *Journal of Quality in Maintenance Engineering*, Vol. 12 No. 3, pp. 239-251.
- Parida, A. and Kumar, U. (2009), *Maintenance Productivity and Performance Measurement. Handbook of Maintenance Management and Engineering*, Springer, London, pp. 17-41.
- Parida, A., Kumar, U., Galar, D. and Stenström, C. (2015), "Performance measurement and management for maintenance: a literature review", *Journal of Quality in Maintenance Engineering*, Vol. 21 No. 1, pp. 2-33.
- Pfeiffer, S. (2017), "The vision of "Industry 4.0" in the making—a case of future told, tamed, and traded", *Nanoethics*, Vol. 11, pp. 107-121.
- Pilone, D. and Pitman, N. (2005), *UML 2.0 in a Nutshell*, O'Reilly Media, Sebastopol, CA.
- Reitemeyer, B. (2020), "Automatic generation of conceptual enterprise models", *2020 IEEE 24th International Enterprise Distributed Object Computing Workshop (EDOCW)*, IEEE, pp. 74-79.
- Roubtsova, E. and Michell, V. (2013), "A method for modeling of KPIs enabling validation of their properties", *Proceedings of the 5th acm sigchi annual international workshop on behaviour modelling-foundations and applications*, pp. 1-10.
- Ruiz, P.A.P., Kamsu-foguem, B. and Noyes, D. (2013), "Knowledge reuse integrating the collaboration from experts in industrial maintenance management", *Knowledge-Based Systems*, Vol. 50, pp. 171-186.
- Ruschel, E., Santos, E.A.P. and Loures, E.D.F.R. (2020), "Establishment of maintenance inspection intervals: an application of process mining techniques in manufacturing", *Journal of Intelligent Manufacturing*, Vol. 31, pp. 53-72.
- Schmiedbauer, O., Maier, H.T. and Biedermann, H. (2020), "Evolution of a lean smart maintenance maturity model towards the new age of industry 4.0", doi: [10.15488/9640](https://doi.org/10.15488/9640).
- Schönig, S., Cabanillas, C., Jablonski, S. and Mendling, J. (2016), "A framework for efficiently mining the organisational perspective of business processes", *Decision Support Systems*, Vol. 89, pp. 87-97.
- Schönig, S., Cabanillas, C., Di Ciccio, C., Jablonski, S. and Mendling, J. (2018), "Mining team compositions for collaborative work in business processes", *Software and Systems Modeling*, Vol. 17, pp. 675-693.
- School, H.B. (2009), *Measuring Performance: Expert Solutions to Everyday Challenges*, Harvard Business Press, Brighton, MA.
- Schwab, K. (2017), *The Fourth Industrial Revolution*, Penguin Books, City of Westminster, London.

-
- Simões, J.M., Gomes, C.F. and Yasin, M.M. (2011), "A literature review of maintenance performance measurement", *Journal of Quality in Maintenance Engineering*, Vol. 17 No. 2, pp. 116-137.
- Sipos, R., Fradkin, D., Moerchen, F. and Wang, Z. (2014), "Log-based predictive maintenance", *Proceedings of the 20th ACM SIGKDD international conference on knowledge discovery and data mining*, pp. 1867-1876.
- SMRP (2017), "Best practices 5th edition Maintenance and reliability body of knowledge".
- Srivastava, N.K., Mondal, S., Chatterjee, N. and Parihar, S. (2018), "Identifying critical factors for various maintenance policies: a study on Indian manufacturing sector", *International Journal of Productivity and Quality Management*, Vol. 25, pp. 41-63.
- Suzuki, T. (1994), *TPM In Process Industries*, Taylor & Francis, Japan Institute of Plant Maintenance, Boca Raton, FL.
- Thaduri, A., Famurewa, S.M., Verma, A.K. and Kumar, U. (2019), *Process Mining for Maintenance Decision Support. System Performance and Management Analytics*, Springer, Singapore, pp. 279-293.
- Uysal, F. and Tosun, Ö. (2012), "Fuzzy TOPSIS-based computerized maintenance management system selection", *Journal of Manufacturing Technology Management*, Vol. 23 No. 2, pp. 212-228.
- Vaidya, S., Ambad, P. and Bhosle, S. (2018), "Industry 4.0—a glimpse", *Procedia Manufacturing*, Vol. 20, pp. 233-238.
- Van Der Aalst, W. (2016), *Data Science in Action. Process Mining*, Springer, Heidelberg, Vol. 2.
- Van Der Aalst, W.M., Reijers, H.A., Weijters, A.J., Van Dongen, B.F., De Medeiros, A.A., Song, M. and Verbeek, H. (2007), "Business process mining: an industrial application", *Information Systems*, Vol. 32, pp. 713-732.
- Wang, J., Li, C., Han, S., Sarkar, S. and Zhou, X. (2017), "Predictive maintenance based on event-log analysis: a case study", *IBM Journal of Research and Development*, Vol. 61, No. 1, pp. 11:121-11:132.
- Wireman, T. (2005), *Developing Performance Indicators for Managing Maintenance*, Industrial Press, New York.
- Wireman, T. (2008), *Successfully Utilizing CMMS/EAM Systems: Maintenance Strategy Series*, Reliabilityweb.com, Fort Myers, FL.
- Wireman, T. (2014), *Benchmarking Best Practices for Maintenance, Reliability and Asset Management*, Industrial Press, New York.
- Wynn, M.T. and Sadiq, S. (2019), "Responsible process mining-a data quality perspective", *International Conference on Business Process Management*, pp. 10-15, Springer.
- Zhai, S., Riess, A. and Reinhart, G. (2019), "Formulation and solution for the predictive maintenance integrated job shop scheduling problem", *2019 IEEE International Conference on Prognostics and Health Management (ICPHM)*, pp. 1-8, IEEE.

Corresponding author

Mehrdad Agha Mohammad Ali Kermani can be contacted at: M_kermani@iust.ac.ir